The role of IT leadership in realizing the return on investment of AI coding assistants.

by

Celumusa H Zulu

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Supervisor: Dr. Siyabonga Mhlongo

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Abstract

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Background

Artificial intelligence (AI) is one of the important structural influencers on productivity growth and competitive advantage today (Gillespie et al., 2025). As of 2023, generative AI systems have progressed from experimentation to operationalization within organizational processes (Gillespie et al., 2025). Recent global studies suggest that organizations are investing more in AI with clear expectations of understanding tangible economic gains, cost savings, and performance improvements (Gillespie et al., 2025). Research on information systems and management has advanced from adoption to value understanding, with a focus on distinguishing organizational value from investment in AI technologies as a function of organizational governance and management capability rather than technological capability (Hossain et al., 2025).

In software engineering, generative AI tools represent one of the most prominent manifestations of AI-enabled augmentation (Peng et al., 2023). Providing developers with functionalities such as code suggestions and entire code functions in real time based on context, documentation, and debugging (Peng et al., 2023).Practical controlled experiments provide conclusive evidence that AI-assisted developers complete software development tasks faster and report increased productivity of about 55.8% compared with control groups (Peng et al., 2023). Collaboration between humans and AI systems is a paradigm shift from the traditional concept of automation, in which AI enhances developers' capabilities rather than replacing them with machines (Peng et al., 2023). Initial performance improvements demonstrated in controlled promising experiments may not necessarily guarantee long-term organizational performance benefits. Empirical studies report considerable variation in the productivity gains from AI-assisted software development tasks across software development teams and organizations.

Both organizational culture and leadership have traditionally been acknowledged as factors that impact the success of technology implementation. In the digital domain, it is the values and norms that foster a culture of experimentation, learning, and data-driven decision-making, as well as a willingness to adopt technological changes. New studies have found that digital culture reinforces the association between human AI competencies and innovation outcomes, suggesting that cultural readiness moderates the performance effects of AI investments (An et al., 2024). Likewise, the same is true for employee engagement in AI initiatives, ethical governance, and leadership's role in influencing strategic alignment. While the importance of digital culture and leadership is increasingly acknowledged in the AI adoption domain, the current body of research remains dispersed (An et al., 2024). Most studies analyse AI adoption from a macro-organizational perspective or focus on individual productivity effects, failing to consider the integration of digital culture, leadership, and ROI in software engineering (Peng et al., 2023). ROI is increasingly becoming the sole rationale for investing in enterprise AI initiatives.

African organizations are accelerating the adoption of AI technologies to address structural inefficiencies and resource limitations. Emerging market organizations have reported high levels of workplace AI experimentation and positive attitudes toward AI productivity (Gillespie et al., 2025). However, the extant literature also emphasizes the lack of digital skills pipelines, governance maturity, and leadership ability, among other things, as difficulties associated with the adoption of GenAI technologies (Hughes et al., 2026). In this regard, the importance of organizational culture and leadership cannot be overstated. In the absence of strategic direction, trust, and a learning-oriented culture, the adoption of AI technologies is more inclined toward fragmentation and symbolic adoption than toward creating value. The digital transformation landscape in Africa is a valid environment for exploring the impact of socio-organizational factors on the achievement of returns from AI technologies.

In the context of software engineering in South Africa, it provides a relevant setting for understanding how organizations embrace and use advanced AI tools. Research indicates that AI is emerging as a key factor influencing organizational change and shaping work practices and culture as organizations increasingly embrace AI tools across their business processes and operations (Murire, 2024). There are opportunities and challenges to address to leverage organizational performance through AI adoption and alignment with organizational culture and structure, as indicated by research reviews on AI tool adoption in organizations (Romeo & Lacko, 2025). Though the adoption of AI technology affects many aspects of the organization's performance positively, like productivity and output, the effect of AI technology on aspects like the organization's culture and employees' morale and engagement is complicated and depends on many factors, as indicated by the study on the impact of AI tools in workplace settings (Kassa & Worku, 2025). Studies on the effect of generative AI tools on software engineering have indicated that AI technology positively affects organizations by enhancing productivity in coding and other software engineering activities through employees' adoption of AI tools (Alenezi & Akour, 2025).

Leadership and organizational culture play an important role in the interpretation, adoption, and usage of AI technologies by work teams. Research on AI adoption has found that AI-related technology adoption is also dependent on leadership styles that facilitate embedding technology initiatives into organizational culture (Romeo & Lacko, 2025). Research on the adoption of AI-related technology has also found that the adoption of AI-related technology is dependent on the adoption of the organizational culture that is conducive to the adoption of the technology (Kassa & Worku, 2025). Organizational culture research further indicates that cultural factors, specifically norms related to innovation, collaboration, and adaptability, modulate how employees engage with AI tools and integrate them into their work (Murire, 2024). In software engineering teams, where AI coding assistants alter routines and task assignments, leadership and culture therefore likely determine the extent to which AI investments yield returns in productivity, quality, and financial outcomes, making them essential variables in understanding realized returns on investment.

Problem Statement

Generative AI has shifted its focus from experiment deployment to integrating it into the core knowledge work process. Substantial experimental results show that generative AI can significantly improve the productivity and quality of cognitive work outcomes, especially among lower-performing members, thereby affecting the performance distribution within the organization (Noy & Zhang, 2023). Field experiment results show that the application of generative AI in the real world can improve the organization's productivity and influence relevant outcomes, such as task completion and retention, indicating that the ROI on generative AI comes from its application within the organization, not the technology itself (Brynjolfsson et al., 2025). In the software development field, three large-scale field experiments have shown that providing developers with GitHub Copilot can improve task completion and development activities, providing empirical support for the ROI of generative AI in coding assistance (Cui et al., 2026). Controlled experiment results show that developers using AI coding assistance complete programming tasks more quickly and at a higher rate, providing a business case for AI coding assistance tools based on productivity (Peng et al., 2023).

Despite these measurable improvements, existing research evidence consistently shows that the performance impact differs across contexts. Research in software engineering has shown that AI-assisted code generation produced by GitHub Copilot improves speed but also creates variability in correctness, maintainability, and review effort; this shows the value to depend on the management and integration of such tools and systems (Moradi Dakhel et al., 2023). Research on AI governance in organizations has argued that to sustain value, organizations must have structured processes and systems to ensure clear accountability and supervisory leadership to transform experimentation into sustained performance improvement (Mäntymäki et al., 2022). Research on organizational innovation with generative AI has also shown that organizations can obtain different performance impacts depending on whether AI is used for automation or augmentation; this shows the role of strategic leadership and organizational culture in determining the benefits obtained (Holmström & Carroll, 2025). Overall, this existing research evidence shows the role of organizational culture and strategic leadership in determining whether the improvements in productivity due to AI coding assistants can be sustained and converted to measurable return on investment.

The relevance of these mechanisms can be further accentuated in the context of Africa since organizations face structural limitations in skills development, governance, and digital sophistication. An empirical study of AI readiness in Africa identified the level of preparedness and the institutional ability to adopt AI, thus emphasizing the role of organizations in the transition from AI pilot projects to performance results (Isagah & Ben Dhaou, 2024). In the context of South Africa, empirical studies of the financial sector have identified the role of organizational factors such as leadership support and organizational readiness in the adoption of new and emerging technologies, thus emphasizing the relevance of these factors to the achievement of ROI in AI coding assistants in software development teams (Matsepe & Van der Lingen, 2022). Empirical research carried out within the context of South Africa suggests the use of sociotechnical systems in AI adoption, as well as the importance of change management for the accomplishment of performance outcomes related to AI adoption (Smit et al., 2024). Another empirical research within the context of South Africa reveals the influence of managerial perceptions regarding AI adoption on the accomplishment of performance outcomes within the context of human resource management as a field of AI application.

Collectively, the global evidence base shows that AI coding assistants can improve productivity, and the organizational evidence base shows that sustained ROI depends on alignment, effective governance, and culture. The evidence from Africa and South Africa also shows that context plays an important role in the success of technology adoption. Therefore, the evidence base provides justification for an in-depth investigation to explore the role of organizational culture and leadership in the achievement of ROI with AI coding assistants in software development teams in South Africa.

Research Objectives

0.1 Main Research Objective

To develop an integrated framework linking financial metrics, organizational culture, and leadership to assess and enhance the return on investment of AI coding assistants in South African software engineering teams.

0.1.1 Sub Research Objectives

1. To identify and operationalize the financial costs and benefits that define return on investment for AI coding assistants.

2. To examine how organizational culture attributes influence productivity and quality outcomes from AI coding assistants.

3. To assess which leadership practices most strongly affect return on investment outcomes.

4. To apply and validate an integrated return on investment framework within large South African organizations.

5. To formulate practical recommendations that enable managers to maximize return on investment through cultural and leadership interventions.

Research Question(s)

0.2 Main Research Question

How do organizational culture and leadership shape the return on investment of AI coding assistants in South African software engineering teams?

0.2.1 Sub Research Questions

1. What financial costs and benefits define return on investment for AI coding assistants in software engineering teams?

2. How do organizational culture attributes influence productivity and quality outcomes from AI coding assistants?

3. Which of the leadership behaviours strongly affect return on investment outcomes from AI coding assistants?

4. How can financial and organizational factors be integrated into a practical return on investment framework for large South African organizations?

5. What practical guidance can be offered to managers to maximize returns from AI coding assistance investments through cultural and leadership interventions?

Keywords: AI, ROI, GenAi.

Significance/rationale

Generative AI has the potential to revolutionize the field of software engineering by increasing productivity as well as the quality of the output in tasks related to knowledge work (Brynjolfsson et al., 2025; Peng et al., 2023). Coding assistants using AI improve the speed as well as the probability of completing tasks by developers. However, the relationship between the productivity of using AI coding assistants and the ROI of software engineering teams is not well understood.

Leadership and organizational culture have been identified as factors that affect the successful adoption of the associated benefits of using AI technology. It affects innovation through guiding organizational resources towards innovative practices (Hossain et al., 2025). A culture of learning, cooperation, and innovation is essential for organizational digital capability (Abdul Wahab & Radmehr, 2024; Li et al., 2025; Poisat et al., 2024). A system perspective, adaptability, is essential for organizational assimilation of digital technologies. Emerging markets' digital performance is dependent on their level of readiness and leadership context (Gillespie et al., 2025; Nurlia et al., 2023).

However, a link is still missing as the literature on AI coding assistants does not show a link between organizational ROI and AI coding assistants' productivity. Similarly, the literature on AI value does not show a link between organizational ROI and AI value. The literature on organizational culture and leadership does not show a link between organizational ROI and AI coding assistants' value for software engineering teams. This study closes this gap as it explores the influence of organizational culture and leadership on ROI realization from AI coding assistants for software engineering teams.

Scope/Delimitation

This study is guided by a quantitative research approach that seeks to explore the impact of organizational culture and leadership on the achievement of return on investment (ROI) for AI coding assistants in software engineering teams. The study is limited to the usage of generative AI tools that facilitate the inclusion of code, inline suggestions, debugging, and documentation, as they are part of the software development process. Literature has already proven that the usage of such tools increases productivity as well as the efficiency of task completion (Brynjolfsson et al., 2025; Noy & Zhang, 2023; Peng et al., 2023).

The target population is software engineering teams working for medium-to-large-scale organizations in South Africa who have prior experience working with AI coding assistants for more than six months. This is a reasonable time frame for evaluating the outcomes since the adoption of AI coding assistants is dependent on firm-level characteristics (Babina et al., 2024). Moreover, the digital readiness of the African continent is recognized as an environmental factor (Gillespie et al., 2025).

The researcher has decided to keep the scope of the independent variables limited to organizational culture and leadership, as they might impact the achievement of return on investment for AI coding assistants. The organizational culture is analysed by taking the learning orientation, collaboration, and innovation support as factors (Abdul Wahab & Radmehr, 2024; Fosso Wamba et al., 2024). Leadership is analysed by taking the strategic direction, governance, and digital support as factors (An et al., 2024; Hossain et al., 2025; Murire, 2024). The return on investment is analysed by taking the quantified performance indicators as factors.

Delimitation

Because the technical efficiency of AI tools has already been studied in experiments (Peng et al., 2023), it is important to highlight that this research does not focus on the accuracy benchmarking, technical algorithmic evaluation of the efficiency of the AI coding assistants, architecture of the AI model, or other technical evaluations of the efficiency. While in this research, there was a choice to focus on the socio-organizational factors of value realization in the context of AI. The research only includes software engineering teams, not other professional groups using generative AI, such as marketing, finance, or legal teams. This is because the software engineering workflow is significantly different from other professions.

The importance of organizational culture and leadership was emphasized in the research as the main independent variables. Other variables that have the potential to impact ROI have not been studied in detail, such as national regulations, the macro environment, and pricing models. Though these variables have a significant impact on ROI, the aim of the research was to concentrate on the internal variables.

Brief Limitations

There are some limitations that are built into the study. Firstly, the calculation of the ROI can depend in part on self-reported perceptions of productivity and quality. Secondly, the study's focus on South African organizations limits the study's statistical generalizability to other geographic regions. Lastly, the study's cross-sectional or short-term longitudinal design may limit the capacity to generalize to long-term financial outcomes because the impact of organizational culture and leadership on performance can develop gradually over time (An et al., 2024).

Key concepts / Definitions

Table 1: Your Table Caption Go Here

Term	**Definition**
AI	Artificial Intelligence
ROI	Return on Investment
SEM	Structural equation modelling

Theoretical / Conceptual framework

This study is based on the following premise: while the productivity gains derived from the use of AI coding assistants can be significant for developers, the value that can be derived from the productivity gains for the organization itself remains an open question. This view is consistent with the value realization framework, which argues that business value depends on how the organization manages, supports, and integrates the technology into its day-to-day business processes. This view also has parallels with the sociotechnical systems view, which argues that the value generated using AI coding assistants depends on the individuals, processes, leaders, and organizational structures that come together.

There is a significant body of research that suggests that generative AI has a positive impact on individual performance. Research conducted in the field has shown that generative AI has a positive effect on the productivity and quality of performance (Brynjolfsson et al., 2025). Experimental research has also shown that generative AI has a positive effect on the efficiency and quality of performance (Noy & Zhang, 2023). Research has shown that programmers working with AI assistants perform their tasks more efficiently and with better quality than those programmers who do not use AI assistants (Peng et al., 2023). However, research conducted at the firm level has shown that the ROI generated by AI is firm-specific, depending upon how the firm invests in AI (Babina et al., 2024).

In this context, leadership and organizational culture are identified as key factors that explain the disparities identified. Leadership is responsible for guiding and directing the firm, as well as the way AI coding assistants are used within the firm. The empirical evidence suggests that leadership is a key factor that affects digital or AI performance outcomes (Hossain et al., 2025; Murire, 2024). On the other hand, organizational culture is responsible for influencing the behavioural aspects of individuals within the firm. Learning culture is a culture that promotes the use of AI tools within the firm through collaboration and innovation. The empirical evidence suggests that a digital organizational culture strengthens the relationship between AI performance outcomes and organizational performance (Fosso Wamba et al., 2024). Furthermore, a digital organizational culture increases innovation outcomes by human-AI capabilities (An et al., 2024).

Accordingly, the conceptual model proposes that ROI from AI coding assistants used within software engineering teams is the key outcome variable, while leadership and organizational culture are identified as the key drivers of this outcome. The way the AI tool is integrated into the organization's workflows and standardized is the conduit variable linking leadership and organizational culture to the key outcome variable of ROI from AI coding assistants. Organizational readiness is included as a control variable because of the differing digital performance outcomes across studies (Gillespie et al., 2025; Nurlia et al., 2023). Accordingly, while AI tools improve individual performance outcomes significantly, the achievement of associated financial returns depends on leadership and organizational culture.

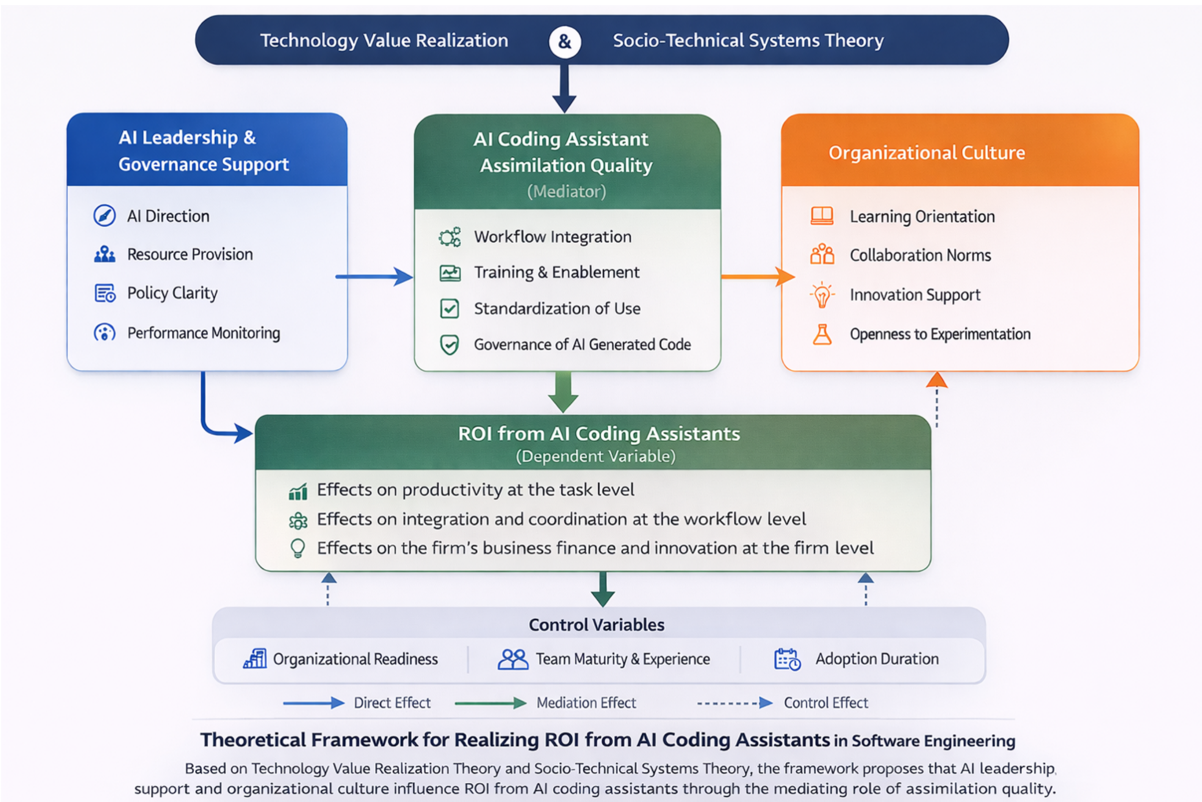

Figure 1: theoretical framework.

Declaration

I, Celumusa H Zulu, certify that the dissertation submitted by me for the Master of Commerce in Information Technology Management degree at the University of Johannesburg is my independent work and has not been submitted by me for a degree at another university.

Celumusa H Zulu

(No signature required)

Dedication

To my Mother,

who always believe in me,

and to all my Siblings

who always believe in me

Acknowledgements

First and foremost, infinite gratitude and appreciation to my Creator, Who has given me the will to start studying again and the endurance to see this through in a time when my world was turned on its head. My perseverance, accomplishments and very being are only by Your will.

You are the very foundation my success is built on.

My deepest gratitude to my Mother, Sister and my brother's as well for always remembering me in your prayers.

My sincerest appreciation to my Supervisor Dr. Siyabonga Mhlongo, who has always been my light through this journey. You got me started on this, thank you for seeing me through to the finish line.

I'd also like to extend my sincerest appreciation to all the people that took their precious time and completed the survey to make this study possible.

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Chapter 1

Introduction

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1.1 Section Heading

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1.1.1 Subsection Heading

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1.1.2 Some maths

LaTeX is very good at presenting mathematics.

Inline equation : $ax^2 + bx + c = 0$

Display equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Numbered equation:

$$\frac{\partial}{\partial t} U + \nabla \cdot F = 0 \tag{1.1}$$

Aligned equation (not how equals signs line up):

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \sum_{i=1}^{n} a_i b_i \\ &= a_1 b_1 + a_2 b_2 + \cdots + a_n b_n; \end{aligned} \tag{1.2}$$

Aligned equation with no numbers:

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \sum_{i=1}^{n} a_i b_i \\ &= a_1 b_1 + a_2 b_2 + \cdots + a_n b_n; \end{aligned}$$

1.1.3 Theorems, proofs, definitions and examples

Theorem 1.1 (Fermat's last theorem)**.** *No three positive integers a, b, and c satisfy the equation $a^n + b^n = c^n$ for any integer value of n greater than 2.*

Proof. Left as an exercise for the reader. □

Definition 1.1. *The intersection of two sets A and B, denoted by $A \cap B$, is the set of all objects that are members of both the sets A and B.*

Example 1.1

Given the two sets $A = \{x : x \in \mathbb{N}, x < 5\}$ and $B = \{x : x \in \mathbb{N}, x \text{ is even}\}$ then $A \cap B = \{2, 4\}$.

1.2 Figures and tables

Figure 1.1: This is a figure caption, note how it appears underneath the figure.

Table 1.1: This is a table caption, not how it appears above the table.

first column	second column	third column
This column is left aligned	This column is centrally aligned	This column is right aligned

1.2.1 Program code

Listing 1.1: A MATLAB function to compute the first n numbers of the Fibonacci series

```
function y = fibonacci(n)

% This function calculates the first n terms in the Fibonacci series

y(1) = 0;
y(2) = 1;

for i = 3 : n
    y(i) = y(i-2) + y(i-1)
end

end
```

1.3 Referencing

For additional information on citation commands, consult the `biblatex` cheat sheet via this URL: https://tug.ctan.org/info/biblatex-cheatsheet/biblatex-cheatsheet.pdf. The comprehensive `biblatex-apa` archive can be found here: https://ctan.org/pkg/biblatex-apa.

Cross referencing is easily done by using the label of the item you are referencing to (see source code for details).

- Chapter 1
- Section 1.1
- Equation (1.1)
- Table 1.1
- Figure 1.1
- Page 2

Above is an example of a bulleted list. You can also create numbered lists:

1. First list item
2. Second list item
   (a) First sub list item
   (b) Second sub list item
3. Third list item

1.4 List of Acronyms

Acronyms can be introduced using the `\gls{}` command. This works in conjunction with what is included in the `acronyms.tex` file. For example, my simple acronym (MSA) is an entry in this file, and MSA can be used throughout the document. See https://www.overleaf.com/learn/latex/Glossaries for more usage details.

Chapter 2

Literature Review

2.1 Introduction

Generative AI is a structural inflection point in knowledge-intensive work. It is different from other forms of AI and automation because the former primarily focuses on complex cognitive work through the creation of outputs that are sensitive to context and require interpretation, refinement, and integration. In the field of software engineering, AI coding assistants in the form of generative code completion tools are embedded into the coding environment and allow for the generation, debugging, documentation, and refactoring of code. Empirical studies of the impact of generative AI on human productivity and output quality revealed that, under conditions of structured task design, a significant increase in human output is achieved. In a large-scale experimental design, the availability of generative AI tools resulted in a reduction in task completion time and an increase in output quality, as determined by the researcher. The results achieved were statistically significant for the total sample, while the results were more pronounced for low-performing individuals (Noy & Zhang, 2023). The results support the idea that generative AI tools result in micro-level productivity effects. However, the productivity gains at the task level do not necessarily translate into organizational return on investment. Research on AI investment at the firm level indicates that the performance effects are heterogeneous and conditional. AI investment is linked with growth through innovation and product development channels; however, the level and stability of the returns depend on the capabilities of the organization (Babina et al., 2024). The market reaction to AI investment announcements is mixed, indicating the level of awareness among investors about the risk and capabilities (Lui et al., 2021).

These studies all circle back to the same basic idea: AI capability does not produce value; rather, value is produced when the technological tools are embedded in the organizational systems. More recent studies have suggested another component: a digital organizational culture can influence the innovation outcome of AI capability (An et al., 2024). Leadership can also play a part in influencing the development of trust and embedding AI usage (Schneider et al., 2023).

Despite the recent increase in research on the performance effects of AI, little research has appeared that combines the effects of organizational culture and leadership into a structured ROI model for AI coding assistants in software development teams. This chapter synthesizes global empirical research on AI productivity and firm performance, critically evaluates the impact of organizational conditions, and focuses the research question more specifically on the South African market, where empirical ROI research has yet to be conducted.

2.2 Generative AI and Productivity: From Task Performance to Organisational Outcomes

The best evidence for how generative AI changes work comes from tightly controlled experiments. In the case of Noy and Zhang (2023), the evidence provides strong causal proof of how gen AI increases the rate of task completion and enhances the quality of professional writing. From this, we can see precisely how the presence of gen AI changes the gap in performance. The benefits of using gen AI are not equally distributed, with the less skilled benefiting the most. This indicates the ability of gen AI to level the playing field for teams.

This uneven benefit to different individuals can, therefore, have implications for software development teams. If the benefits to gen AI for coding tasks are greater for the less skilled, the composition of the team might change. However, the benefits to the team must not only be seen at the task level. They must be sustained, integrated, and monetized to increase the ROI.

This distinction finds corroboration in firm-level evidence. Babina et al. (2024) show a positive association between the adoption of AI and firm growth and innovation, with the Association is stronger for firms with complementary capabilities. The channel through which this association seems to occur is mainly through product innovation rather than labour substitution effects, implying the importance of AI in value creation through product development rather than cost reduction.

At the same time, there is evidence of mixed reactions in the financial markets to AI investment announcements. Lui et al. (2021) show that some firms have positive abnormal returns around the announcement of AI investments, whereas others have neutral or negative returns. This appears to reflect investors' ability to distinguish between firms with the ability to successfully implement AI and those with a risk of implementation failure.

In the context of software engineering, empirical evidence points to the fact that there is a great deal of usage of AI coding assistants. However, this usage is done in an inconsistent manner. In a comprehensive empirical study done to examine the experiences of software developers, it was found that there was usage of AI coding assistants for various activities in the software development process. However, issues of trust, boundaries, and governance ambiguity have been found to hinder the effective impact of this usage (Sergeyuk et al., 2025). This points to the fact that the frequency of usage alone may not guarantee the achievement of benefits.

According to the literature, the return on investment (ROI) model of AI coding assistants comprises a three-tier structure:

Chapter 3

Research Methodology

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Chapter 4

Results

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Chapter 5

Discussion

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Chapter 6

Conclusion

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Appendix A

Ethical Clearance Certificate

UNIVERSITY OF JOHANNESBURG

COLLEGE OF BUSINESS AND ECONOMICS RESEARCH ETHICS COMMITTEE (CBEREC)

Dear CH Zulu,

ETHICAL APPROVAL GRANTED FOR RESEARCH PROJECT

Decision: Clearance granted

This letter serves to confirm that the proposed research project indicated in the table below, has been reviewed by the School of Consumer Intelligence and Information Systems (SCiiSREC) Research Ethics Committee of the University of Johannesburg (UJ). Ethical clearance is hereby granted and is valid for three years, from 1 May 2026 until 30 April 2029.

Applicant	CH Zulu
Supervisor/s	Dr S Mhlongo
Student/staff number	226166283
Title	The role of IT leadership in realising the return on investment of AI coding assistants
Decision date at meeting	30 April 2026
Reviewers	SCIISREC 2/9 RV1 V2 AISREC Dr F Radebe; Dr S Dube
Ethical clearance code	2026SCiiS024
Rating of application	CODE 01
CODE 01 – Approved	CODE 02 – Approved with suggestions/requirements with no re-submission
CODE 03 – Referred back	CODE 04 – Disapproved, cannot re-submit

The researcher/s may now commence with the study providing that:

1. Researcher/s, i.e. internal and/or external applicants, who wish to undertake research at the University of Johannesburg (UJ) case institution as data collection source involving UJ students and/or UJ staff and/or any UJ support services submit the Personal Information Impact Assessment (PIIA) via the 2026 UJ Gatekeeper Platform, after familiarising themselves with the UJ POPIA and PAIA manual and, if applicable, utilising the UJ login to register POPI requests.

2. The researcher/s understand the implications of any form of data breach and takes responsibility for any possible violation or any form of unethical research conduct. The onus is on the researcher/s to ensure that the project adheres to ethical research requirements.

3. The researcher/s will be conducting the study as set out in the approval application, i.e. Form_7a and/or Form_7b linked to the departmentally approved research proposal that was submitted by the researcher/s, together with supporting documents, primary data collection, and/or secondary data analysis, to the research ethics committee.

4. The researcher/s will ensure that the project adheres to all applicable legislation, scopes of practice, professional codes of conduct, and scientific standards as it pertains to the field or study.

5. The researcher/s will bring under the attention of the research ethics committee any proposed changes, concerns that arise, and/or unexpected ethical management issues.

6. Any changes that can affect the study-related risks for participants or researchers must be reported immediately to the research ethics committee in writing.

7. No data collection or fieldwork activities may continue after the ethical clearance has expired. A request for an extension of ethical clearance can be made in writing to the research ethics committee.

8. The researcher/s must be aware that infectious diseases may influence researcher/s, research participants, and responses. Steps must be taken to the best of researcher/s' ability to avoid the spread of infectious diseases.

9. Researcher/s familiarise themselves with the PAIA and POPIA UJ compliance manual. All data collection must be done within Government regulations, for instance, researcher/s must uphold all provisions of the Protection of Personal Information Act (POPIA) Act 4 of 2013 when collecting, processing, and storing data. This includes but is not limited to upholding the principles of accountability, processing restriction, purpose, transparency, processing restrictions, security, and rights of participants.

10. Researcher/s abide by the ethical use of Artificial Intelligence (AI). As non-legal entities, AI tools cannot assert the presence or absence of conflicts of interest nor manage copyright and license agreements. UJ requires researcher/s to disclose any use of AI in the research process and the preparation of their research report to ensure transparency and allow for proper assessment by reviewers and examiners.

The School of Consumer Intelligence and Information Systems Research Ethics Committee wish the researcher/s all the best with this research project.

Kind regards,

T du Plessis

Prof T du Plessis
SCiiSREC Chair
tduplessis@uj.ac.za

Appendix B

Appendix Chapter

B.1 Appendix section

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Appendix C

Another Appendix Chapter

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